

信号与系统

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and Information Engineering

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State variable analysis of the system

- ※ Basic concepts and general forms of system state variable analysis
- ※ Establishment of state equation and output equation for continuous time system
- ※ Establishment of state equations and output equations for discrete-time systems
- ※ The solution of state equation and output equation of continuous time system
- ※ The solution of state equation and output equation of discrete-time system



Establishment of state equation and output equation for continuous time system

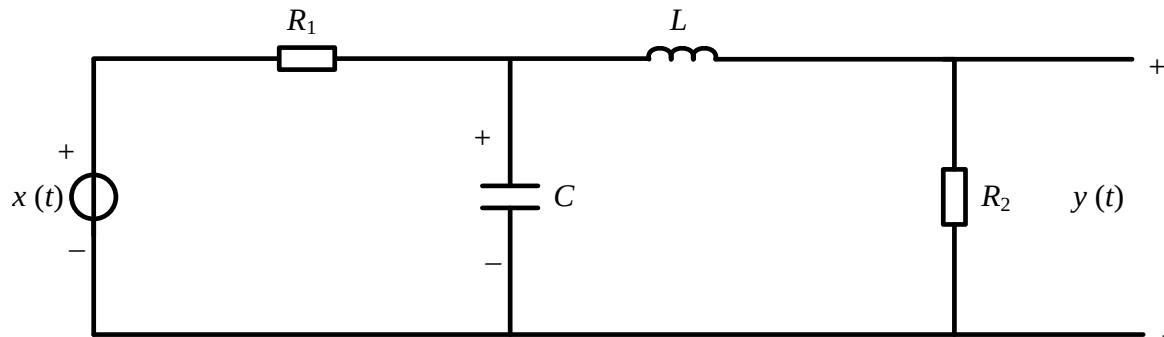
Continuous systems are often described in the form of circuits, differential equations, system functions or analog block diagrams, from which state equations and output equations of continuous systems are established.

- ✘ The state equation and output equation are established by the circuit
- ✘ State equations and output equations are established from differential equations
- ✘ State equation and output equation are established by simulating block diagram
- ✘ State equations and output equations are established by system functions



The state equation and output equation are established by the **circuit**

[Example] The state equation and output equation of the following circuit are established.

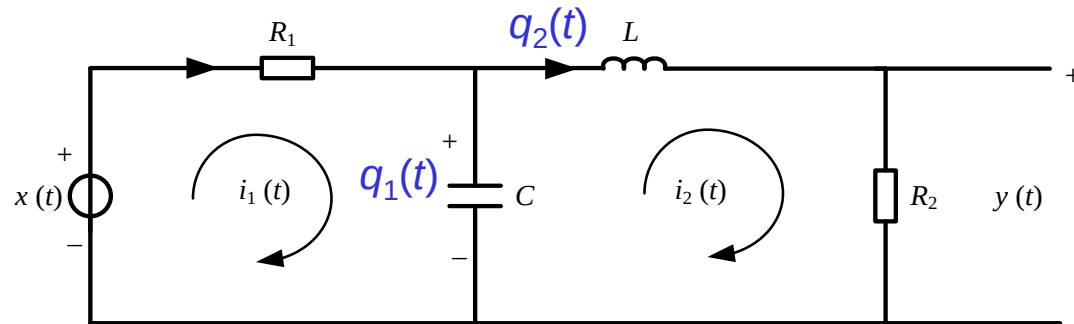


Solution:

There are two independent dynamic components in the circuit, so two state variables are required.



The state equation and output equation are established by the **circuit**



Capacitive voltage $v_c(t)$ and inductor current $i_L(t)$ are selected as state variables $q_1(t)$ and $q_2(t)$.

$$q_2(t) = i_2(t) \quad C\dot{q}_1(t) = i_1(t) - i_2(t)$$

Loop equation:

$$R_1 i_1(t) + q_1(t) = x(t)$$

$$L\dot{i}_2(t) + R_2 i_2(t) - q_1(t) = 0$$



The state equation and output equation are established by the **circuit**

State equation
$$\begin{cases} \dot{q}_1(t) = -\frac{1}{R_1 C} q_1(t) - \frac{1}{C} q_2(t) + \frac{1}{R_1 C} x(t) \\ \dot{q}_2(t) = \frac{1}{L} q_1(t) - \frac{R_2}{L} q_2(t) \end{cases}$$

Output equation
$$y(t) = R_2 q_2(t)$$

Written in matrix form:

State equation
$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1 C} & -\frac{1}{C} \\ \frac{1}{L} & -\frac{R_2}{L} \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1 C} \\ 0 \end{bmatrix} x(t)$$

Output equation
$$y(t) = \begin{bmatrix} 0 & R_2 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix}$$



The state equation and output equation are established by the **circuit**

- (1) Inductance current and capacitance voltage are generally selected as state variables;
- (2) Loop voltage equation around the derivative of inductance current;
- (3) The derivative column node current equation around the capacitor voltage;
- (4) After sorting out steps (2) and (3), the equation of state can be obtained;
- (5) Solve the relationship between output and state variables and input, and get the output equation.。



State equations and output equations are established from differential equations

[Example] Given the differential equation describing a system, try to write its state equation and output equation.

$$y''(t) + 3y'(t) + 2y(t) = x(t)$$

Solution: Select $y(t)$ and $y'(t)$ as the state variables $q_1(t)$ and $q_2(t)$ of the system,

$$q_1(t) = y(t) \quad q_2(t) = y'(t)$$

The equation of state of the system obtained from the differential equation is

$$\begin{cases} \dot{q}_1(t) = q_2(t) \\ \dot{q}_2(t) = y''(t) = x(t) - 2q_1(t) - 3q_2(t) \end{cases}$$

The output equation is

$$y(t) = q_1(t)$$



State equations and output equations are established from differential equations

The matrix form of the equation of state is

$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} x(t)$$

The matrix form of the output equation is

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix}$$



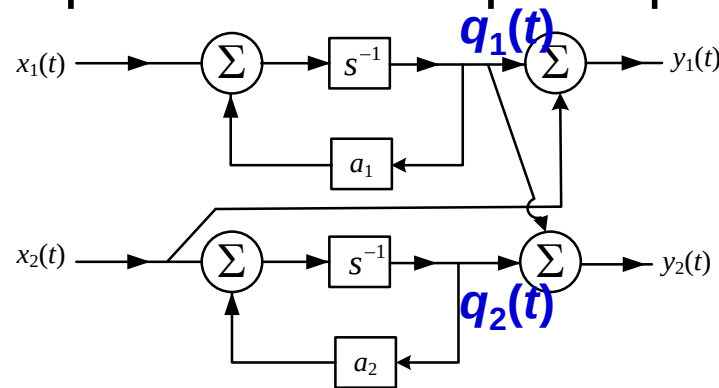
State equations and output equations are established from differential equations

- (1) Selected $y(t)$, $y'(t)$, $y''(t)$,... as the state variables of the system $q_1(t)$, $q_2(t)$, $q_3(t)$,... The number of state variables for a differential equation of order n is n .
- (2) Establish state equation and output equation according to differential equation.



State equation and output equation are established by simulating **block diagram**

[Example] Given the simulation block diagram of a continuous system, its state equation and output equation are established.



Solution: Select the **output end** of the integrator as the state variables $q_1(t)$ and $q_2(t)$.

List the **equation of state** around the **integrator input**:

$$\dot{q}_1(t) = x_1(t) + a_1 q_1(t) \quad \dot{q}_2(t) = x_2(t) + a_2 q_2(t)$$

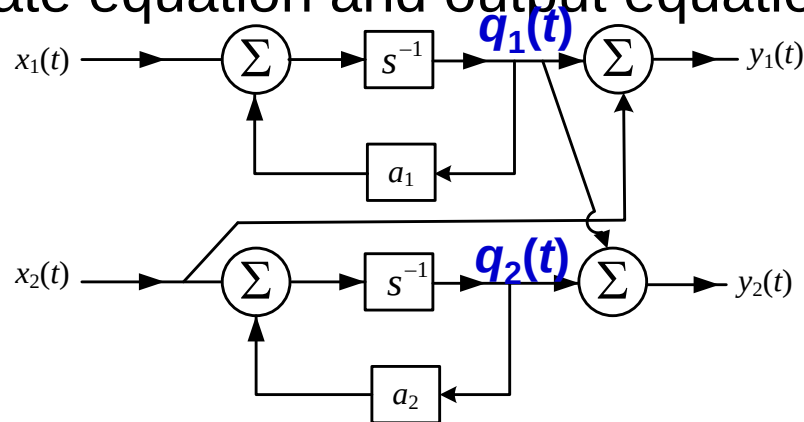
List the **output equation** based on the **continuous system output**:

$$y_1(t) = q_1(t) + x_2(t) \quad y_2(t) = q_1(t) + q_2(t)$$



State equation and output equation are established by simulating **block diagram**

[Example] The simulation block diagram of a continuous system is given, and its state equation and output equation are established



$$\dot{q}_1(t) = x_1(t) + a_1 q_1(t)$$

$$\dot{q}_2(t) = x_2(t) + a_2 q_2(t)$$

$$y_1(t) = q_1(t) + x_2(t)$$

$$y_2(t) = q_1(t) + q_2(t)$$

Solution: **State equation**

$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Output equation

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$



State equation and output equation are established by simulating **block diagram**

- (1) Select **the output end of the integrator** as the state variable;
- (2) Write the equation of state around **the input column of the integrator**;
- (3) Write the output equation around **the output column of the continuous system**.



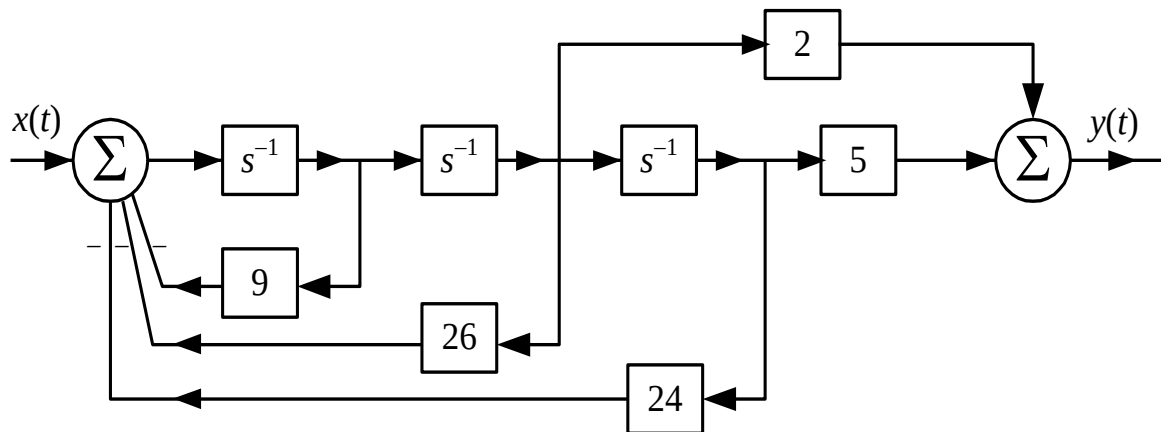
State equations and output equations are established by **system functions**

[Example] It is known that the system function describing a continuous system is

$$H(s) = \frac{2s + 5}{s^3 + 9s^2 + 26s + 24}$$

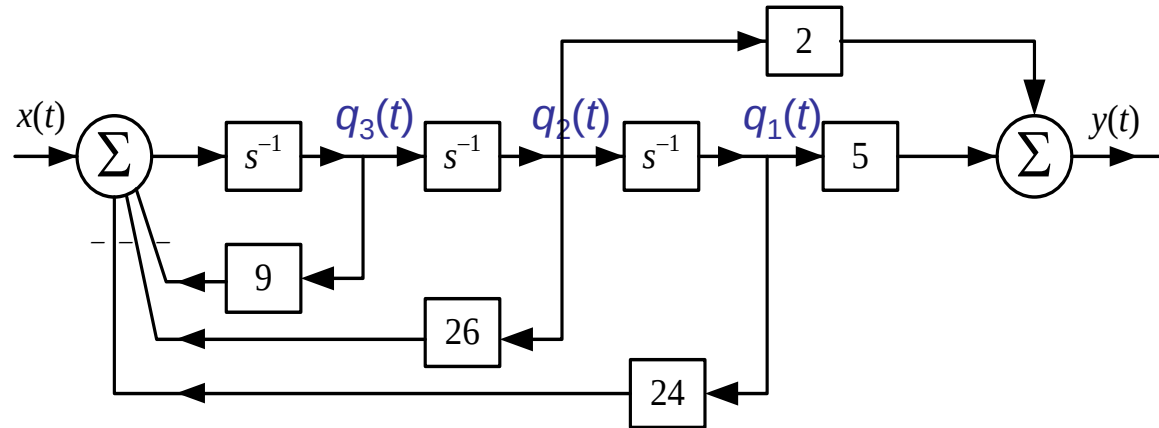
The simulation block diagrams of **direct, cascading and parallel** structures of the system are drawn, and the state equations and output equations are established.

Solution: (1) Direct block diagram





State equations and output equations are established by **system functions**



The **integrator outputs** the **state variables** $q_1(t)$, $q_2(t)$ and $q_3(t)$ of the system, with

$$\dot{q}_1(t) = q_2(t) \quad \dot{q}_2(t) = q_3(t)$$

$$\dot{q}_3(t) = -24q_1(t) - 26q_2(t) - 9q_3(t) + x(t)$$

$$y(t) = 5q_1(t) + 2q_2(t)$$



State equations and output equations are established by **system functions**

$$\dot{q}_1(t) = q_2(t) \quad \dot{q}_2(t) = q_3(t)$$

$$\dot{q}_3(t) = -24q_1(t) - 26q_2(t) - 9q_3(t) + x(t)$$

$$y(t) = 5q_1(t) + 2q_2(t)$$

Matrix form of **the equation of state**:

$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} x(t)$$

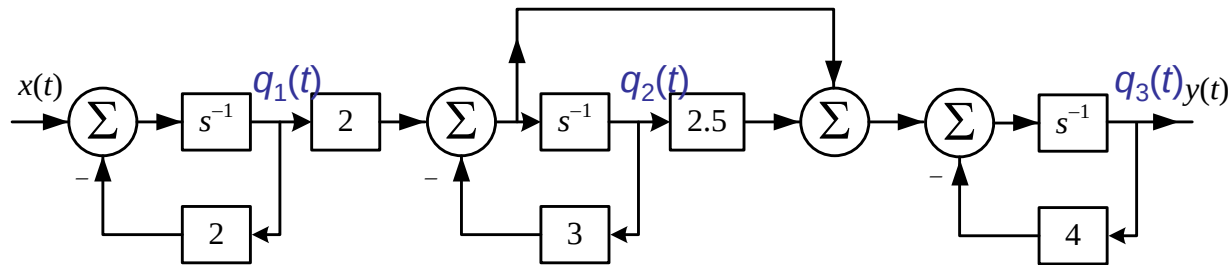
Matrix form of **the output equation**:

$$y(t) = [5 \ 2 \ 0] \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix}$$



State equations and output equations are established by **system functions**

(2) Cascading block diagram $H(s) = \left(\frac{2}{s+2} \right) \left(\frac{s+2.5}{s+3} \right) \left(\frac{1}{s+4} \right)$



Select **the output end of the integrator** as the **system state variables** $q_1(t)$, $q_2(t)$ and $q_3(t)$,

$$\dot{q}_1(t) = -2q_1(t) + x(t)$$

$$\dot{q}_2(t) = 2q_1(t) - 3q_2(t)$$

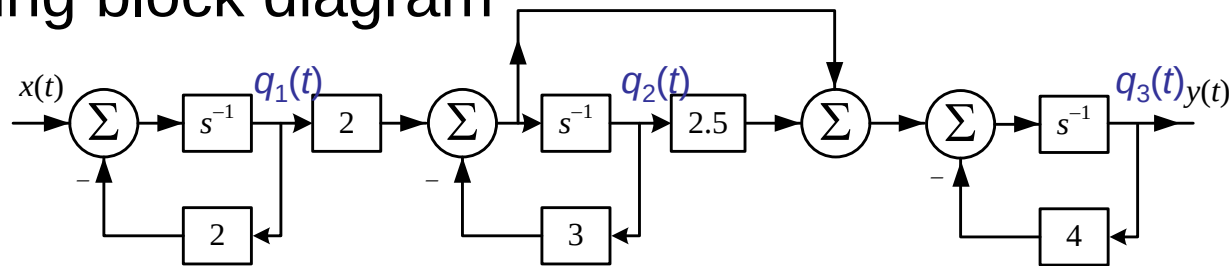
$$\dot{q}_3(t) = 2.5q_2(t) + \dot{q}_2(t) - 4q_3(t) = 2q_1(t) - 0.5q_2(t) - 4q_3(t)$$

$$y(t) = q_3(t)$$



State equations and output equations are established by **system functions**

(2) Cascading block diagram



Matrix form of **the equation of state**

$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 2 & -3 & 0 \\ 2 & -0.5 & -4 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} x(t)$$

Output equation in matrix form

$$y(t) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix}$$



State equations and output equations are established by **system functions**

(3) Parallel block diagram

$$H(s) = \frac{0.5}{s+2} + \frac{1}{s+3} - \frac{1.5}{s+4}$$

State equation

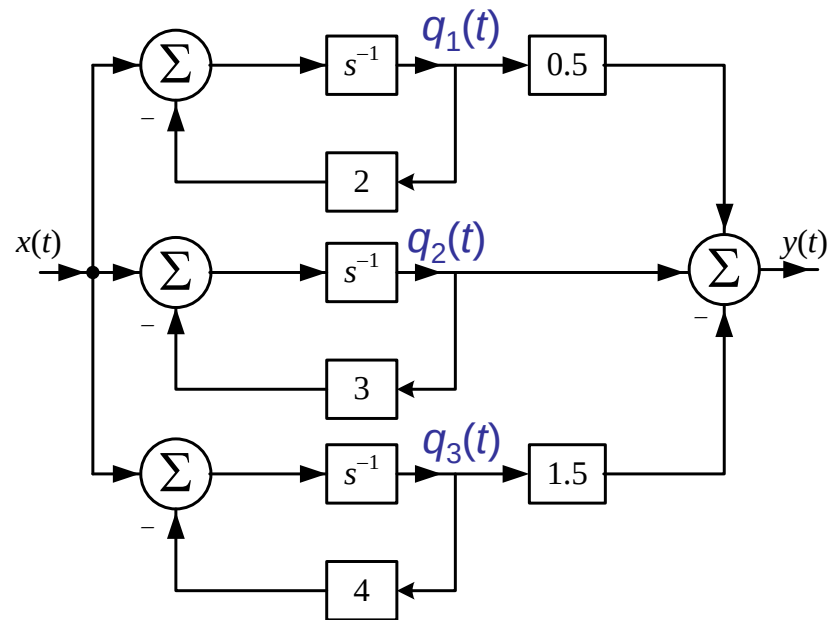
$$\dot{q}_1(t) = -2q_1(t) + x(t)$$

$$\dot{q}_2(t) = -3q_2(t) + x(t)$$

$$\dot{q}_3(t) = -4q_3(t) + x(t)$$

Output equation

$$y(t) = 0.5q_1(t) + q_2(t) - 1.5q_3(t)$$





State equations and output equations are established by **system functions**

(3) Parallel block diagram

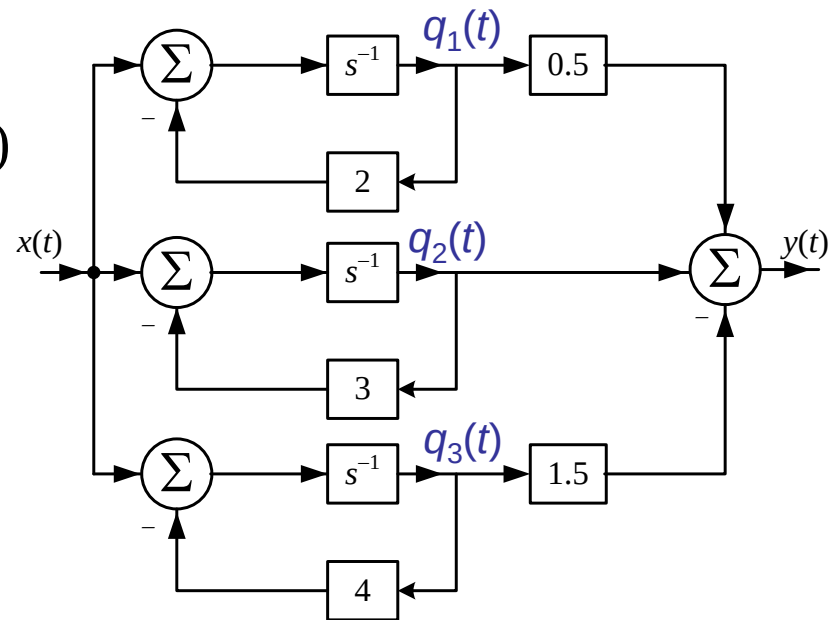
$$H(s) = \frac{0.5}{s+2} + \frac{1}{s+3} - \frac{1.5}{s+4}$$

Matrix form of the equation of state

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} x(t)$$

Output equation in matrix form

$$y(t) = \begin{bmatrix} 0.5 & 1 & -1.5 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix}$$





State equations and output equations are established by **system functions**

The simulation block diagram of continuous system is drawn by **system function**.

The state equation and output equation of the system are established by the simulation **block diagram**.

The equation of state and the output equation are different in the simulation block diagram of different structures.



Thanks

Some of the materials quoted in this course are accumulated by the main lecturer over the years, from a variety of media and colleagues, peers and friends of the exchange, it is difficult to attribute them all, so I would like to state and thank you!